

frequency of 312.5MHz. By using different numbers of bits at each level, multiple speeds can be obtained depending on channel conditions. This results in a flexible scheme that can adapt to many POF channels.

Transmission of data is structured in frames that are sent continuously. Each frame contains 112 data blocks (MLCC code words) and also additional information related to the physical layer. This information is in the form of two pilot signals (S1 and S2) and the physical header (PHS). Pilots are intended to aid the receiver in performing linear and non-linear equalisation and timing recovery.

Non-linear equalisation is implemented in the receiver while linear equalisation is implemented in part in the transmitter using Tomlinson-Harashima Precoding (THP) to avoid error-propagation issues, and in the receiver using a feed-forward equaliser. The physical header is used for physical-layer signaling, which includes selection of MLCC configuration and transmission of THP coefficients. GigaPOF scheme also provides good energy efficiency.

The Ethernet moves packets through networks on a best-effort, non-deterministic basis, which makes it poorly suited for use in mission-critical real-time systems used to control drivetrain or braking systems or support driver-assist functions. But later developments have added mechanisms like IEEE 802.1tsn (time-sensitive networking) protocol that provide assured, non-conflicting delivery of data and command packets within strict time parameters.

Additional standards for specific timing-critical applications are also available, including IEEE 802.1AS/802.1Q/802.1Qca protocols, which support transport, switching and management of latency-sensitive connections used in high-speed real-time control applications.

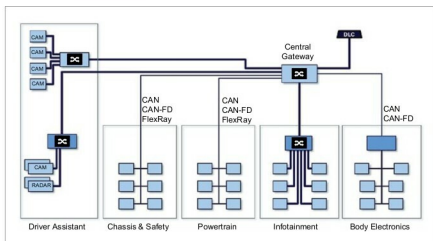


Fig. 5: Driver-assistance and infotainment systems

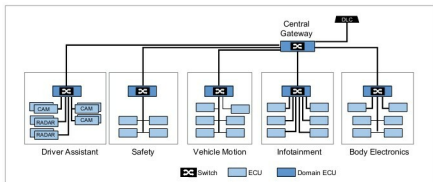


Fig. 6: Ethernet in automobiles (Source: Bosch)

The next generation of infotainment systems will benefit from IEEE 802.1AVB, which enables aggregation and distribution of audio/video streams. These capabilities rely on IEEE 1588 precision-timing protocol to provide timing information required by each node to keep its clock synchronised with the network's master clock. IEEE 1588 is now providing precision-timing services in everything from small-cell wireless networks to industrial automation systems.

Various car makers currently use the Ethernet to transport video streams within their infotainment systems. But the Ethernet must cross the reliability gap from these applications, where a bus failure means an inconvenient loss of cabin comforts to being certified for safety-critical applications such as drivetrain, suspension or braking

systems where failure is not even an option.

Bridging allows the Ethernet to peacefully co-exist with legacy sub-systems; even more as IEEE 802.1BASE-T1 and 1000BASE-T1 standards developments bring technologies like the two-wire Ethernet PHYs to the industry.

Other protocols like IEEE 802.1TSN Deterministic Ethernet standard will also play an important role, bringing precision, reliability and speed that the Ethernet needs to handle mission-critical automotive systems like advanced driver-assist and collision-avoidance systems and, ultimately, the car's main system bus.

Many driver-assistance systems rely on getting data from cameras and other sensors in a timely manner. Unlike watching video on a computer, where buffering is used